

# Noah Weidig

## GIS Analyst & Data Scientist

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GIS & Remote Sensing Research Associate leveraging remote sensing, GIS, and data science to translate complex geospatial data into clear insight about how our world changes. Focused on wildland-urban interface wildfire risk, land-use change, and spatial decision-making.

## Education

### Master of Science, Forest Resources & Conservation

Aug 2023 – Aug 2025

University of Florida · Gainesville, FL

Thesis: Large Wildfire Patterns in the WUI · 4.0 GPA

### Bachelor of Science, Biological Sciences

Aug 2018 – May 2022

Northern Kentucky University · Highland Heights, KY

Ecology, Evolution, & Organismal Track · 4.0 GPA

## Experience

### GIS & Remote Sensing Research Associate

Aug 2025 – Present

University of Florida · Milton, FL

Led GIS-based analysis assessing spatial patterns and land-use impacts on community and regional risk management.

### Graduate Research Assistant

Aug 2023 – Aug 2025

University of Florida · Milton, FL

Conducted geospatial analysis to support wildland-urban interface and emergency management decision-making.

### Research Assistant Intern

May 2023 – Aug 2023

USDA Agricultural Research Service · Miles City, MT

Collected, cleaned, and validated spatial datasets to support GIS-based risk assessment and land-use planning.

### Fire & Recreation Intern

Jan 2023 – May 2023

Student Conservation Association / US Forest Service · Wise, VA

Applied GIS to support spatial planning and risk assessment for land management projects.

### Research Assistant

May 2022 – Jan 2023

University of Cincinnati · Cincinnati, OH

Conducted quantitative data analysis for biomedical research using advanced statistical modeling and visualization.

### CAD Design Specialist

Sep 2021 – Jun 2022

Garage Living · Cincinnati, OH

Developed CAD drawings and project plans for residential remodeling projects.

### Research Assistant

Feb 2020 – May 2022

Northern Kentucky University · Highland Heights, KY

Collected, organized, and analyzed complex ecological datasets using spatial and statistical methods.

### Elections Data Specialist

Aug 2017 – Feb 2020

Boone County Clerk's Office · Burlington, KY

Managed and analyzed precinct-level GIS data for voter registration and electoral planning.

## Publications

### JOURNAL ARTICLES

**Fueling the fire: predicting cogongrass (*Imperata cylindrica*) impacts on fire behavior across environmental and seasonal gradients**

Ivory II, A. A., Wonkka, C. L., McGranahan, D. A., Akpejelu, P., Kincy, S., **Weidig, N. C.**, Roberts, C. P., Brock, K. M., Silva, C. A. & Donovan, V. M. · *Fire Ecology*, vol. 22, no. 1, pp. 119 (2026)

**Regional differences in infilling and land-use conversion characterize woody cover increases across the Eastern United States**

Ivey, M. A., **Weidig, N. C.**, Ivory II, A. A. & Donovan, V. M. · *Forest Ecology and Management*, vol. 603, pp. 123446 (2026)

**Woody cover fuels large wildfire risk in the eastern US**

Ivey, M. A., Wonkka, C. L., **Weidig, N. C.** & Donovan, V. M. · *Geophysical Research Letters*, vol. 51, no. e2024GL110586 (2024)

Changing large wildfire dynamics in the wildland–urban interface of the eastern United States

**Weidig, N. C.**, Wonkka, C. L., Ivey, M. A. & Donovan, V. M. · *International Journal of Wildland Fire*, vol. 33, no. 12 (2024)

The Goldilocks principle: Finding the balance between water volume and nutrients for ovipositing *Culex* mosquitoes (Diptera: Culicidae)

**Weidig, N. C.**, Miller, A. L. & Parker, A. T. · *PLOS ONE*, vol. 17, no. 11, pp. e0277237 (2022)

## THESIS

Patterns and drivers of large wildfires within and surrounding the wildland-urban interface of the eastern United States

**Weidig, N. C.** · *University of Florida* (2025)

## Presentations

Fire at the fringe: Drivers of large wildfires in and near the wildland-urban interface of the eastern United States

**Weidig, N. C.**, Wonkka, C. L., Ivey, M. A., Johnson, D. J. & Donovan, V. M. · *Ecological Society of America* (2025)

Shifting forest structure and composition following decades of fire exclusion in the eastern U.S.: Implications for prescribed fire restoration and wildfires

Alexander, H. D., Donovan, V. M., Lamounier Moura, A., Lazzaro, L. G., **Weidig, N. C.** & Ivey, M. A. · *Proceedings of the 4th fire ecology and management symposium (FES-412)*, vol. 4 (2025)

Unravelling changing wildfire regime dynamics in the eastern United States

Donovan, V. M., **Weidig, N. C.**, Ivey, M. A., Crandall, R. M., Fill, J. M. & Wonkka, C. L. · *International Association for Landscape Ecology - North America Annual Conference* (2025)

Woody cover increases linked to forest infilling and land conversion in the eastern United States

Ivey, M. A., **Weidig, N. C.**, Ivory II, A. A. & Donovan, V. M. · *International Association for Landscape Ecology - North America Annual Conference* (2025)

Southeastern *Myotis* (*Myotis austroriparius*) Use of Stormwater Sewer Systems

Ivory II, A. A., Brock, K. M., Johnson, S. A., **Weidig, N. C.**, Hallett, M. T., Acevedo, M. A. & Scheffers, B. R. · *Florida Bat Working Group Annual Meeting; Southeastern Bat Diversity Network* (2025)

Key drivers of large wildfires in and near the eastern United States wildland-urban interface

**Weidig, N. C.**, Wonkka, C. L., Ivey, M. A., Johnson, D. J. & Donovan, V. M. · *University of Florida Office of Graduate Professional Development Graduate Student Research Day* (2025)

#### Changing Large Wildfire Patterns in the Eastern United States Wildland-Urban Interface

**Weidig, N. C.**, Wonkka, C. L., Ivey, M. A. & Donovan, V. M. · *Northwest Florida Association of Environmental Practitioners Symposium* (2024)

#### R Markdown: A Reproducible Workflow

**Weidig, N. C.** · *University of Florida West Florida Research and Education Center Invited Seminar* (2024)

#### Large Wildfire Patterns in the Wildland-Urban Interface of the Eastern U.S.

**Weidig, N. C.**, Wonkka, C. L., Ivey, M. A. & Donovan, V. M. · *International Association for Landscape Ecology - North America Annual Conference* (2024)

#### Changing Large Wildfire Risk in the Southeastern U.S.

Donovan, V. M., Crandall, R. M., Fill, J. M., Ivey, M. A., **Weidig, N. C.** & Wonkka, C. L. · *Gulf of Mexico Conference* (2024)

#### Rumble in the Cogongrass Jungle: Shrub Resiliency from Round 1

Carl, N., Shows, C., McKeithen, J., **Weidig, N. C.** & Donovan, V. M. · *Gulf of Mexico Conference* (2024)

#### Inflammation mediated reprogramming of platelets following transcatheter aortic valve replacement

**Weidig, N. C.**, Niang, D., Williams, M., Porema, K., Palumbo, J., Weyrich, A., Smyth, S. & Lynch, D. R. · *International Society for Applied Cardiovascular Biology Biennial Meeting* (2022)

#### The Goldilocks principle: Finding the balance between water volume and nutrients for ovipositing *Culex* mosquitoes (Diptera: Culicidae)

**Weidig, N. C.**, Miller, A. L. & Parker, A. T. · *Entomological Society of America Annual Conference* (2021)

## Webinars

#### Burning Boundaries: Large Wildfire Patterns and Drivers in the Eastern United States Wildland-Urban Interface

**Weidig, N. C.**, Wonkka, C. L., Ivey, M. A., Johnson, D. J. & Donovan, V. M. · *Southern Fire Exchange* (2025)

#### Changing large wildfire regime dynamics in the eastern United States

Donovan, V. M., Crandall, R. M., Fill, J. M., Ivey, M. A., **Weidig, N. C.** & Wonkka, C. L. · *Oak Woodlands & Forests Fire Consortium* (2025)

### Increasing Large Wildfires in the Eastern United States

Donovan, V. M., Crandall, R. M., Fill, J. M., Ivey, M. A., **Weidig, N. C.** & Wonkka, C. L. · *Southern Fire Exchange* (2024)

## Media coverage

### Georgia blaze shows how climate change has led to more wildfires in the East

Borenstein, S. · *Associated Press* (2026)

### Changing Large Wildfire Dynamics in the Wildland–Urban Interface of the Eastern United States: Research Brief

Fernandez, C. & **Weidig, N. C.** · *Fire Lines*, vol. 15, no. 7 (2025)

### Woody Cover Fuels Large Wildfire Risk in the Eastern U.S.

Ivey, M. A., Wonkka, C. L., **Weidig, N. C.** & Donovan, V. M. · *Oak Woodlands & Forests Fire Consortium* (2025)

### Wildfire surges in East, Southeast US fueled by new trees and shrubs

Lester, L. · *AGU News* (2024)

## Service

### PEER REVIEW

Referee report. For: Integrated Fire Management and Closer to Nature Forest Management at the Landscape Scale as a Holistic Approach to Foster Forest Resilience to Wildfires [version 3; peer review: 4 approved with reservations, 2 not approved]

Donovan, V. M. & **Weidig, N. C.** (2025)

## Awards & honors

### Outstanding Thesis in Natural Resources

University of Florida, College of Agricultural & Life Sciences · 2025 Academic Year

2026

### Outstanding Thesis in Forest Resources & Conservation

University of Florida, School of Forest, Fisheries, & Geomatic Sciences · 2025 Academic Year

2026

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|--------------------------------------------------------------------------------------------------|------|
| <b>Best Student Presentation</b>                                                                 | 2025 |
| International Association of Landscape Ecology - North America · \$300                           |      |
| <b>BioLEAPS Travel Grant</b>                                                                     | 2025 |
| International Association of Landscape Ecology - North America · \$1,650                         |      |
| <b>Student Travel Grant</b>                                                                      | 2025 |
| International Association of Landscape Ecology - North America · \$850                           |      |
| <b>Outstanding Teaching Assistant of the Year</b>                                                | 2025 |
| University of Florida, School of Forest, Fisheries, & Geomatic Sciences · 2024 Academic Year     |      |
| <b>IFAS Travel Grant</b>                                                                         | 2024 |
| University of Florida, Institute of Food & Agricultural Sciences · \$250                         |      |
| <b>Doris and Earl Lowe and Verna Lowe Scholarship</b>                                            | 2024 |
| University of Florida, College of Agricultural & Life Sciences · \$2,000                         |      |
| <b>Graduate Research Assistantship</b>                                                           | 2023 |
| University of Florida, College of Agricultural & Life Sciences · \$27,500 annually for two years |      |
| <b>Outstanding Graduate in Biology</b>                                                           | 2022 |
| Northern Kentucky University, Department of Biological Sciences                                  |      |
| <b>1st Place, Undergraduate Student Poster Competition</b>                                       | 2021 |
| Entomological Society of America · \$75                                                          |      |
| <b>Dr. Carol Swarts Milburn Research Fellowship</b>                                              | 2020 |
| Northern Kentucky University · \$3,500                                                           |      |
| <b>President's List Awardee</b>                                                                  | 2018 |
| Northern Kentucky University · 4.00 GPA · 2018–2022                                              |      |

## Skills

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|----------------------|----------------------------------------------------------------------------------|
| LANGUAGES            | R, Python, JavaScript, SQL                                                       |
| MARKUP               | Markdown, LaTeX, HTML, CSS                                                       |
| LIBRARIES            | tidyverse, pandas, NumPy, Plotly                                                 |
| GIS & REMOTE SENSING | ArcGIS Pro, Google Earth Engine, QGIS, GDAL, sf, terra, GeoPandas, OpenStreetMap |
| WORKFLOWS            | Git, GitHub, GitHub Actions, Conda, renv                                         |